

ReactJS HOL-11

Objective:

Create a React application demonstrating event handling, event handlers, synthetic events, and a currency converter component.

Theory

Explain React Events

React events are actions triggered by user interactions such as clicks, key presses, and form submissions. React handles these events using event handlers.

Explain Event Handlers

Event handlers are JavaScript functions that execute in response to user actions. They are attached using JSX attributes such as `onClick` and `onChange`.

Define Synthetic Event

A Synthetic Event is React's cross-browser wrapper around the native browser event, providing consistent behavior across browsers.

React Event Naming Convention

React event names use camelCase, such as `onClick`, `onSubmit`, `onChange`, and `onKeyDown`, instead of lowercase HTML event names.

Prerequisites

- Node.js
- npm
- Visual Studio Code

Implementation Steps

1. Create a React project named `eventexamplesapp`.
2. Create buttons for Increment, Decrement, Say Welcome and OnPress.
3. Implement multiple methods for the Increment button.
4. Create a `CurrencyConvertor` component.
5. Convert Indian Rupees to Euro on button click.
6. Run the application using `npm start`.

App.js

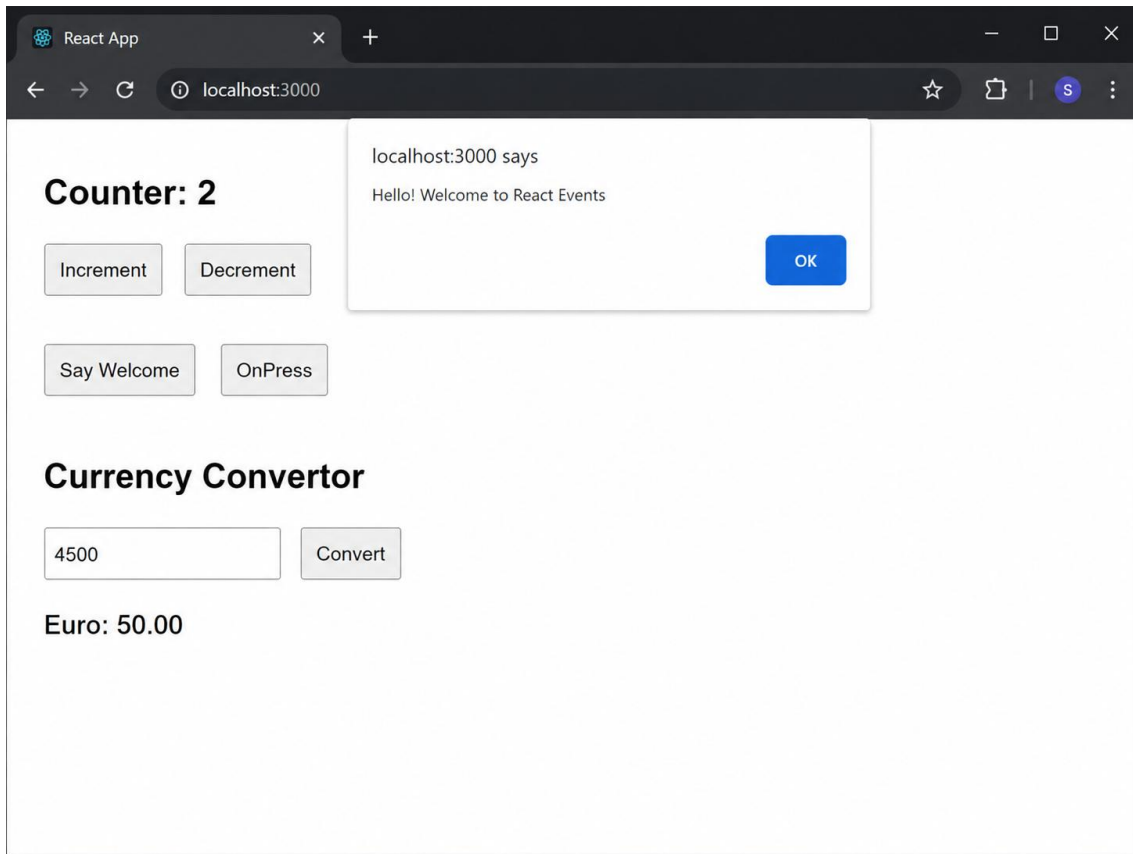
```
import React,{useState} from 'react';
import CurrencyConvertor from './CurrencyConvertor';
function App(){
  const [count,setCount]=useState(0);
  const hello=()=>alert('Hello! Welcome to React Events');
  const inc=()=>{setCount(count+1);hello();};
  const dec=()=>setCount(count-1);
  const welcome=(msg)=>alert(msg);
  const press=()=>alert('I was clicked');
  return(<div>
    <h2>Counter: {count}</h2>
    <button onClick={inc}>Increment</button>
    <button onClick={dec}>Decrement</button><br/><br/>
    <button onClick={()=>welcome('Welcome')}>Say Welcome</button>
    <button onClick={press}>OnPress</button>
    <CurrencyConvertor/>
  </div>);
}
export default App;
```

CurrencyConvertor.js

```
import React,{useState} from 'react';
export default function CurrencyConvertor(){
  const [rupees,setRupees]=useState('');
  const [euro,setEuro]=useState('');
  const handleSubmit=()=>setEuro((rupees/90).toFixed(2));
  return(<div><h2>Currency Convertor</h2>
    <input value={rupees} onChange={e=>setRupees(e.target.value)} placeholder='INR' />
    <button onClick={handleSubmit}>Convert</button>
    <p>Euro: {euro}</p></div>);
}
```

Output Screenshot:

Increment and Decrement update the counter. Increment also displays a greeting. 'Say Welcome' shows a welcome message. 'OnPress' displays 'I was clicked'. CurrencyConvertor converts Indian Rupees to Euro when the Convert button is clicked.



Learning Outcome

Implemented React event handling, synthetic events, multiple event methods, and a simple currency converter.